

Application of Structural and Topological Features to Recognize Online Handwritten Bangla Characters

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This article presents a set of novel features for robust online Bangla handwritten character recognition. Two feature extraction methods are presented here. The first describes the transition from background to foreground pixels and vice versa. The second uses a combination of topological features and centre-of-gravity- (CG) based circular features where global information, local information, and Circular Quadrant Mass Distribution information have been extracted. The impact of each along with their combination have also been analyzed. A total of 15,000 isolated online Bangla character samples have been collected and used for the evaluation. A Support Vector Machine classifier records the best recognition rate when the transition count feature, CG-based circular features, and topological features are combined.

CCS Concepts: • **Computing methodologies** → *Computer vision problems; Object recognition;*

Additional Key Words and Phrases: Online handwriting recognition, bangla script, transition count, cg-based circle, global information, local information, circular quadrant mass distribution

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1 INTRODUCTION

Online handwritten character recognition for different languages has become an important area of research in the past few decades. This area has gained popularity due to the rapid increase in the use of devices such as the A4 Take Note, the iPad, and Smart phones. Such devices are becoming part of our day-to-day life, and these devices can store handwritten data in the form of online information. Individuals can write in their normal style as freely as with pen and paper. Because

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there is still a desire to convert this written form to editable electronic content, researchers are now showing more interest in Online Handwriting Recognition (OHR). Some good research works for Devanagari [11, 22–25, 46, 47, 50], English [1, 3, 4, 15, 49, 52, 55, 56], and Gurumukhi/Gujarati [6, 20, 28, 32–36, 41–45] scripts have already been published. Many authors have adopted cutting-edge machine-learning approaches like deep learning [2, 10, 13, 14, 29, 48, 53, 54] and extreme learning machine [18] for different scripts. By contrast, for Bangla script, researchers still have not much explored the problem, and there are limited research materials available in the literature.

In Reference [31], the authors defined OHR as the problem of recognizing a handwritten input that is realized as a stream of pen positions collected by using either a digitizer or other pen position sensor. Both sequential and dynamic information are extracted from pen movements on the writing pads. These information act as features for the proposed online Bangla Handwriting recognition system. These features are then fed to a quadratic classifier for recognition. The authors of Reference [26] have presented benchmark recognition results for four popular Indic scripts based on two existing feature extraction techniques known as point-float and direction code histograms. Efficiency of their system has been tested using Nearest Neighbor, Hidden Markov Model (HMM), and Multi-Layer Perceptron (MLP) classifiers. In Reference [16], the authors have recognized Bangla word samples by extracting sub-stroke-level features and proposed an HMM for recognition. The authors in Reference [8] have developed a large annotated database consisting of unconstrained online Bangla handwritten characters. A semi-automatic system has been proposed for the annotation of characters at the boundary. Another approach, in Reference [30], extracts component strokes from characters. Sequential and dynamic features extracted at the stroke level are used for stroke recognition. This information is collected by tracking the pen movements on the writing pad. Characters are constructed from the recognized strokes by matching the stroke sequences stored in a database.

Authors have also proposed techniques related to the segmentation and recognition of online Bangla cursive texts that contain basic and compound characters along with all the modifiers [7]. Text is segmented into primitives to be recognized using directional features. These primitives may be basic/compound character or a part of a basic/compound character. The authors of Reference [17] propose an unsupervised feature generation technique based on dissimilarity space and embedded the local neighborhoods around the pen points along the trajectory. The authors in Reference [27] have manually grouped constituent strokes forming characters into 54 classes based on the similarity of shapes in the graphemes. For stroke recognition, they have constructed one HMM for each stroke class. Characters are finally recognized by the classification of stroke results using look up tables. In Reference [9], the authors have reported an approach for handwritten Bangla recognition of basic characters with the help of direction code-based features. The authors in Reference [5] describe a representation at the stroke level based on their structure or shape, where each stroke corresponds to a sequence of shape features. Strokes are then recognized from stroke database using Dynamic Time Warping (DTW). The author of Reference [19] has designed a novel feature vector by considering the direction of writing, curvature, curliness, inclination, and the standard deviation of the x and y coordinates at the stroke level. Strokes are divided into nine local zones for feature extraction, and classification is performed using a Support Vector Machine (SVM) classifier. The authors in Reference [39] have shown the effectiveness of distance-based feature extraction for the recognition of online handwritten Bangla characters. This method segments the character into N segments, and the pairwise distances are calculated for each point. The individual impact of both global and local information is considered in Reference [37] for online handwritten Bangla characters. The global information estimates shape statistics, and valuable regional information is extracted through local information. The authors in Reference [37] have also use the combination of global and local information. The authors in Reference [40] have

Table 1. Basic Bangla Character Set

অ	আ	ই	ঈ	উ	ঊ	ঋ	এ	ঐ	ও
ক	খ	গ	ঘ	ঙ	চ	ছ	জ	ঝ	
ট	ঠ	ড	ঢ	ণ	ত	থ	দ	ধ	
ন	প	ফ	ব	ভ	ম	য	র	ল	শ
ষ	স	হ	ড়	ঢ়	য়	ৎ	ং	ঃ	ঁ

applied Hausdorff Distance (HD) and Directed Hausdorff Distance–(DHD) based features. After performing a quad-tree-based image segmentation, the authors in Reference [38] have reported three different feature extraction techniques along with their all possible combinations for the recognition of online handwritten Bangla characters. Among these three techniques, the first one computes an area feature using composite Simpson’s rule, and the other two procedures compute mass distribution and chord length of the data pixels present in a character image.

After analyzing the above works, we have come to the conclusion that despite some research attempts there is a lot of room for improvement, such as (1) enhancing recognition accuracy, (2) distinguishing similar shaped character symbols, and (3) developing a large dataset. The strong resemblance among some characters deteriorates the performance of recognition system, so features are designed in this article to handle such problems.

The extraction of discriminating features from pre-processed samples of handwritten characters carries the most significant contribution towards any character recognition system. Motivated by this, two different feature extraction approaches are introduced, the transition count feature and the combination of topological feature along with centre-of-gravity–(CG) based circular feature. Transition count features reflect the statistics of switching between background and foreground pixels and vice versa. The topological feature tracks the presence of crossing points within the character. For the CG-based circular feature, three different techniques are discussed. First, global information calculates the distances from CG to all data pixels. Second, sample characters are divided into four regions, and some local information in the form of ratio of background to foreground pixels in each region are estimated. In the third case, the regions are again sub-divided into three small sub-regions and the ratio of background to foreground pixels is computed in each small sub-region. The individual impact of transition count feature vector and combination of topological feature with CG-based circle approaches have been discussed along with the effect of their combination on the said recognition process.

Bangla is the second most popular language in India. It is one of the official languages and scripts in India and the official language in Bangladesh. The script is inherited from the ancient Brahmi script. The writing style is from left to right like other Indo-Aryan scripts, such as Devanagari. The basic character set of Bangla script contains 39 consonants and 11 vowels, which are shown in Table 1 [39].

For the recognition of online Bangla basic characters, the collected data first go through a basic pre-processing step, where smoothing removes noisy and duplicate (or repeating points) points. Then the points of the input character are sampled to obtain a predefined fixed number of equidistant points (λ). Scaling is then performed by normalizing the boundary size. After the pre-processing stage, the features are extracted, and character samples are recognized using well-known classifiers.

This article is structured as follows: Section 2 reflects the pre-processing steps performed before feature extraction. Section 3 discusses the various feature extraction strategies used for the recognition of online characters. Section 4 analyzes the results, and a summary is presented in Section 5.

2 DATABASE PREPARATION WITH PRE-PROCESSING

There is no standard database of online handwritten Bangla characters. So we have built a representative database. Three hundred handwritten specimens of each Bangla character sample have been collected from 100 different people. Different sections of society with varying age groups, educational backgrounds, and genders have contributed to the data collection process. As the Bangla basic alphabet set contains 50 distinct character shapes, the size of the present database becomes 15,000. No burden was imposed on the writers during the data collection drive. *Take Note* was used to collect the data, where each sample is represented as a collection of pen positions p_t , and t ranges from 1 to M . Suppose that p_t describes the position of the pen having x and y coordinates as x_t and y_t on the basis of pen-up or pen-down status. Here, M represents the total number of pen positions for each of the character samples. During the pre-processing step, duplicate or repeated points are removed. If two consecutive pen points are represented by p_i and p_k , respectively, then the i th point p_i is retained with respect to k th point p_k if

$$x^2 + y^2 > m^2,$$

where, $x = x_j - x_k$ and $y = y_j - y_k$. The value of m is set to 0 to remove all redundant points. Next, a new sequence of approximately equidistant points is obtained by re-sampling the pixel points of the input character into λ points that retains the structure of the character. The value of λ has been chosen as 64 after few trial runs. While performing the scaling operation, all normalized points are scaled to a window of size 64×64 .

3 FEATURE EXTRACTION

The feature set used to recognize online handwritten Bangla characters includes Transition Count, CG-based circular (global information, local information, and Circular Quadrant Mass Distribution (CQMD)), and Topological feature.

3.1 Transition Count Feature

This approach converts the character into a binary image of size 64×64 . A pixel is set to “1” if it is a data point (foreground pixel) and “0” otherwise (background pixel). Based on CG, the character image is then divided into four sub-regions. In each of these sub-regions, the transition counts from background to foreground pixels and vice versa are computed along four directions: row, column, and along two major diagonals (i.e., at four different angles $\theta = 0^\circ, 90^\circ, 45^\circ, 135^\circ$). The physical interpretation of traversing the four directions is to estimate the data distribution along different directions. It is to be noted that the size of these four sub-regions depends on the position of the CG. Figure 1(a) illustrates one such scenario. We have observed that maximum value of the transition count for a particular row/column/diagonal within a sub-image of a character varies from 0 to 6. Therefore, to get an idea about how different values of transition count vary over the entire image, we measure the frequencies of these counts and use them as feature values.

For each sub-image, 28 (i.e., 7×4) feature values are generated for “0” to “1” transition and another 28 for “1” to “0”. Thereby, a total of 56 (i.e., 28×2) feature values are generated for each sub-image. Finally, a total of 224 (i.e., 56×4) feature values are produced for the whole image. For easy reference, feature calculation (here, “0” to “1” transition) on a hypothetical image is described in Figure 1(b).

3.2 CG-Based Circular Feature

In this approach, three different feature vectors are computed by following the global information, local information, and CQMD information extraction strategies. In all three cases, a CG-based circle is drawn over the character sample, and different feature values are estimated. Details of these feature computation processes are described in the following sub-sections.

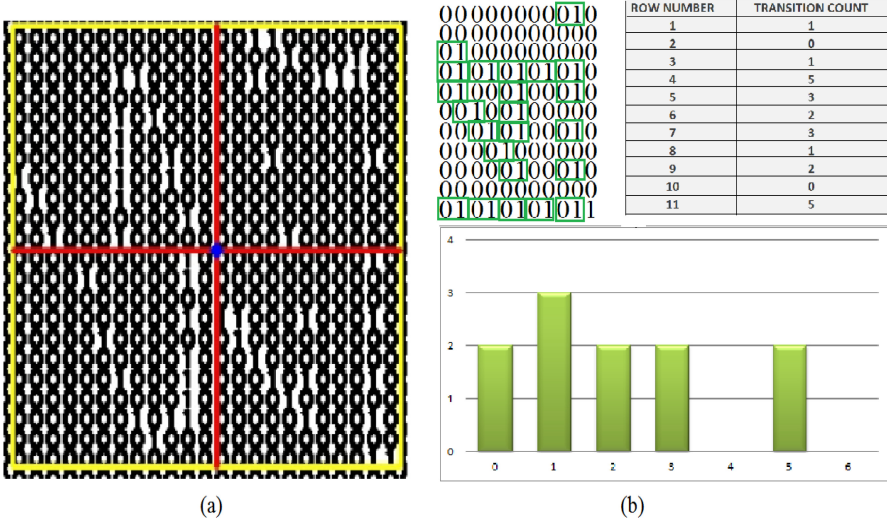


Fig. 1. Transition count feature calculation for character ‘আ’.

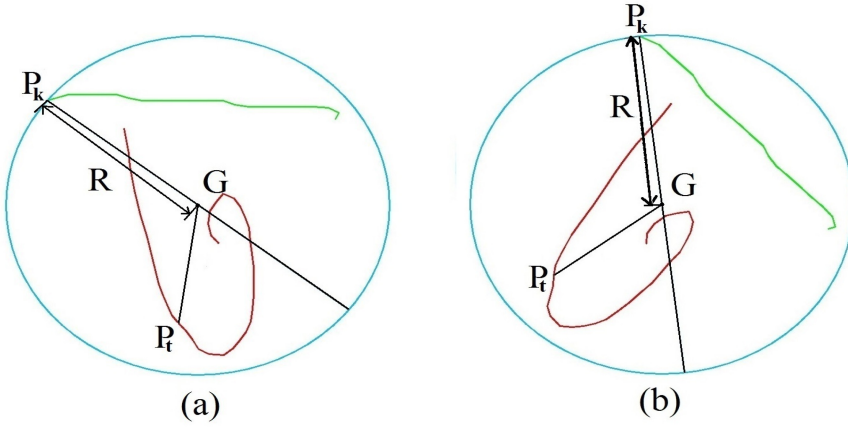


Fig. 2. Global information-based feature calculations for character sample ‘ত’ when the image is rotated (a) 0° and (b) 45° clockwise.

3.2.1 Global Information. To extract global information from an online Bangla character sample, the CG is first estimated, and the distance between the farthest data pixel of the sample from the CG is measured. Considering this distance as a radius, a circle is drawn that completely encloses the character sample. Then, the ratios of the radius of the circle to the distances from the CG of the character to every pixel point P_t are evaluated, where t ranges from 1 to 64 (since all the characters are normalized to 64 points during the pre-processing stage). As an outcome, 64 such ratios are produced by this technique that act as feature values and play an important role in recognizing character samples. After a detailed analysis of the produced results (see Section 4), it can be shown that these feature values have the ability to justify the writing pattern of the characters meticulously. Figure 2(a) and (b) describes the picture when the character is enclosed within the

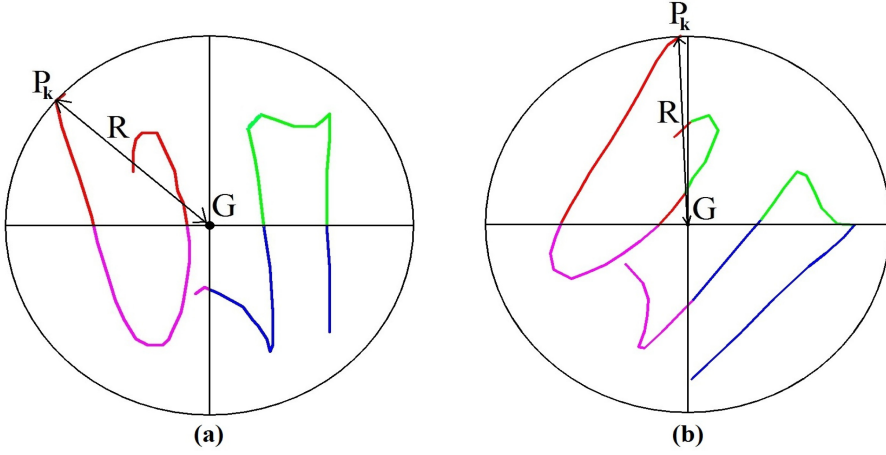


Fig. 3. Extraction of local information-based feature for the character sample 'आ' when rotated (a) 0° and (b) 45° clockwise.

circle after estimating CG and the way to compute all such ratios. In this figure, for an arbitrary pixel point P_t , one ratio calculation is shown, where G denotes the CG, P_k is the farthest pixel from CG, R represents radius of the circle, and P_t is an arbitrary data point of that sample character. The ratio $(P_t G : R)$ is recorded. Next, the character is rotated 45° in the clockwise direction (see Figure 2(b)) and the procedure is repeated to yield 64 more feature values. Hence, considering the rotation, a total of 128 (i.e., 2×64) feature values are produced from each character sample.

3.2.2 Local Information. To extract local features, a CG-based circle is extracted as described in the previous section. The circular region is sub-divided into four sub-regions centered on the CG. This procedure estimates the ratio of foreground pixels (object pixels) to background pixels in each sub-region and considers them as features. Figure 3(a) and (b) depicts the pixel contributions in each sub-region when the sample character is processed and rotated at 0° and 45° in the clockwise direction. Algorithm 1 shows the steps to evaluate local information-based features from a character sample.

This procedure describes the mass distributions in each of the sub-region. As a consequence, we have obtained four feature values from all four sub-regions. The character sample is rotated 45° in the clockwise direction and Algorithm 1 is repeated. Hence, we estimate 8 (i.e., 2×4) feature values as local information.

ALGORITHM 1: Algorithm to compute the CG-based local information feature

STEP 1: START

STEP 2: Calculate CG of the character sample.

STEP 3: Find the distance R from CG to the farthest data pixel from the CG.

STEP 4: Consider this distance R as radius and form a circle over the character sample.

STEP 5: Divide circular region into four sub-regions based on CG.

STEP 6: For each sub-region, find the ratio of the number of data pixels to the background pixels.

STEP 7: END

3.2.3 Circular Quadrant Mass Distribution. In this procedure, a CG-based circle is also formed over the character sample and then the sample is divided into four sub-regions (as mentioned for

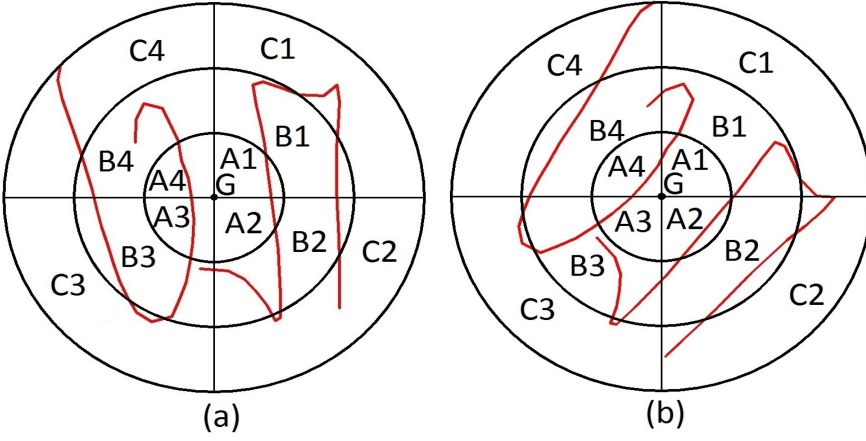


Fig. 4. Illustration of the CQMD feature extraction technique when the character ‘আ’ is rotated (a) 0° and (b) 45° clockwise.

the local information extraction technique). Let R be the radius of the circle. We then draw two more concentric circles with radii as $R/3$ and $2R/3$, respectively, as shown in Figure 4. As a result, each sub-region is again divided into three small sub-parts, marked A_i , B_i , and C_i , where $i = 1, 2, 3, 4$ as shown in Figure 4. In each sub-part of each sub-region, the ratio of foreground to background pixels is computed and used as feature.

Although the basic idea of CQMD is the same as that for the local information feature extraction approach, this technique can estimate finer shape detail. As the writing pattern of different Bangla characters are different and cursive in nature, the pixel distributions inside the sub-parts, A_i , B_i , and C_i , vary significantly. This difference enhances the ability to distinguish different character patterns effectively and thereby augments the efficiency of the recognition process. In this approach, for each sub-region, three ratios are produced (as there are three sub-parts in each sub-region). Hence, a total of $12 (= 4 \times 3)$ feature values are computed by considering all four sub-regions of the corresponding character sample. The character sample is then rotated 45° in a clockwise direction, and the same steps are repeated. Finally, the CQMD approach produces $24 (= 2 \times 12)$ element feature vector from each character sample.

3.2.4 Topological Feature.

• Crossing point

There are a few character pairs in the Bangla alphabet that exhibit strong similarities. For example, ঞ and ঞ are quite similar in structure, except ঞ has a crossing point in the lower left side due to formation of a loop. Hence, the existence of crossing point acts as a good feature to differentiate such character pairs. To compute this feature, a minor modification has been made with the sample character. All the missing pixels between two consecutive data points are estimated. This feature monitors whether a character sample contains a loop. For example, consider Figure 5, where the existence of loop can easily be observed. In this figure, the crossing point is marked green. It is clear that only those pixel points of the sample character are crossing points that have more than three adjacent neighbors and thus help to form a loop. This situation is clearly depicted in Figure 5, where blue points are the neighbors of corresponding crossing point. A total of five feature values (the number of crossing points, the coordinates of the first crossing point, and the coordinates of closest neighbor of the first crossing point) are computed using this procedure.

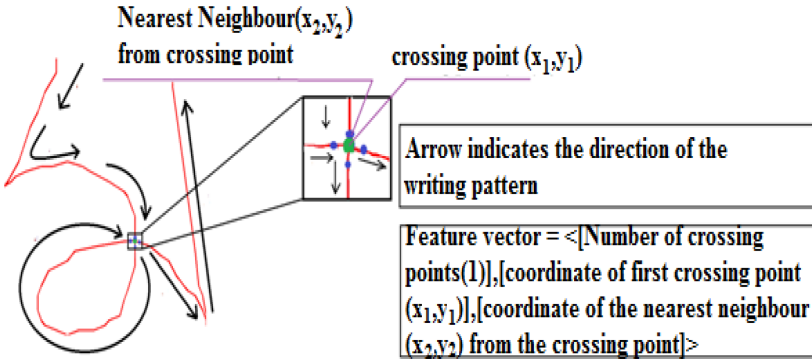


Fig. 5. Identification of a crossing point in character sample 'ঞ'.

Table 2. Illustration of Feature Set and Their Combination Used in the Present Work

SL#	Features Used	Feature length	
1.	Transition Feature	224	
2.	Combination of CG-based circular features	160	
	Topological feature (Crossing Point)	5	
3.	Combination of Transition and CG-based circular features with topological features	Before PCA	389
		After PCA	124

Hence, when all three CG-based circle approaches are combined, then the size of the feature vector becomes 160 (i.e., $128 + 8 + 24$) element feature vector. For topological features, another five feature values are produced. Therefore, a total of 165 ($160+5$) features are generated from each character sample.

4 RESULTS ANALYSIS AND DISCUSSION

In this article, two different feature extraction strategies are introduced: the transition count feature and a CG-based circular feature with a topological feature. In this section, experiments are performed to verify the strength of individual and the combined feature vectors. Table 2 reflects the set of the features used in the current experiment and the dimension of all the feature vectors. These feature vectors are fed to four well-known classifiers [21] viz., SVM, MLP, BayesNet, and Simple Logistic. A fivefold cross validation scheme is applied on a 15,000 character dataset. Table 3 reports the recognition rates observed by the said classifiers. Gray cells represent the maximum recognition rate shown by a classifier. From Table 3, it can be observed that in case of the transition count-based feature, the SVM produces an accuracy of 87.46%, which is the best score among all the classifiers.

Note that the combination of a CG-based circular feature with a topological feature produces a feature vector of length 165 ($160+5$). To reduce the feature dimension, Principle Component

Table 3. Accuracies of Different Classifiers in Recognizing Online Handwritten Bangla Characters

Classifier	Success Rates(%)		
	Transition Count Feature	Global Information + Local Information + CQMD + Crossing Point	Transition Count Feature + Global Information + Local Information + CQMD + Crossing Point
SVM	87.46	93.15	94.56
MLP	69.49	86.34	93.07
BayesNet	70.38	82.47	86.45
SimpleLogistic	82.22	86.42	91.70

Table 4. Accuracy Obtained by Tuning the Parameter γ of SVM Classifier

value of γ	Value of coef0	Database size	Accuracy (in %)
0	1	15,000	95.34
	2		95.47
	3		95.49
	4		95.44

Analysis (PCA) is applied. The resultant feature vector is then evaluated on the said classifiers. It has been observed that in this case also SVM produces the best accuracy of 93.15% . For CG-based circular feature estimation, we have rotated of the character image by 45° to get some valuable information that can be used to recognize skewed or slanted character samples. To validate this, we have tested the feature vector generated by combining topological features and CG-based circular features without any rotation. In this situation, the SVM produces the highest recognition accuracy of 90.34%. Clearly, this accuracy is less than the accuracy obtained when topological feature is combined with a CG-based circular feature considering rotation of the images by 45° .

When the transition count-based feature is added with the combination of the topological feature with a CG-based circular feature, total 389 (224 + 165) features are produced. Again, PCA is carried out and the resultant feature vector (124-element) is fed to the classifiers. At this stage, SVM [51] also produces the best result with 94.56% success rate. Here, we have selected C-SVC type SVM from the LibSVM library and its kernel type is polynomial: $(\gamma \cdot u \cdot v + \text{coef0})^{\text{degree}}$ having $\text{eps}=0.001$, $\text{coef0}=0$ and $\gamma=0.0$ (default value). We have tuned the value of γ and coef0 to see their impact on the system. Changing the γ value does not make significant changes in the final outcome, but when we tune the value of coef0 (0, 1, 2, 3), the performance of the system increases gradually. If we further increase the value of coef0 beyond 3, then the performance of the system degrades. The tuned value of coef0 and the corresponding outcomes are shown in Table 4. From this table, we observe that the recognition accuracy increases up to 95.49% when the value of coef0 is set to 3. Figure 6 shows a graphical representation of success rates achieved for transition count-based features, a combination of topological features with CG-based circle approaches, and the combination of transition count and CG-based circle features with topological features. If

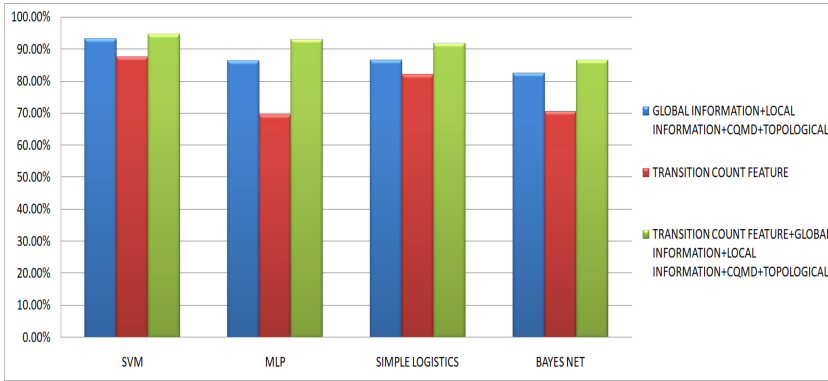


Fig. 6. Graphical representation of the success rates for different classifiers in recognizing online handwritten Bangla basic characters.

Table 5. Recognition Accuracies Attained by Four Different Classifiers on 10 Randomly Selected Datasets for Performing Statistical Significance Test (Numbers in Bracket Indicate the Ranks)

Classifier	Dataset										Mean Rank
	1	2	2	4	5	6	7	8	9	10	
MLP	98.6 (2)	99.3 (2)	98.6 (2)	99.07 (3)	96.07 (3)	97.27 (2)	96.2 (3)	94.93 (3)	72.87 (2)	78.27 (1)	R1=2.3
SVM	99.2 (1)	99.8 (1)	98.8 (1)	99.4 (1.5)	98.27 (1)	98.3 (1)	98.5 (1)	96.8 (1)	73.3 (1)	76.93 (2)	R2=1.15
BayesNet	98.2 (3.5)	98.73 (4)	97.2 (4)	98.13 (4)	91.33 (4)	95 (4)	93 (4)	92.27 (4)	44.07 (4)	47.47 (4)	R3=3.95
Simple Logistic	98.2 (3.5)	99 (3)	98.47 (3)	99.4 (1.5)	96.4 (2)	96.87 (3)	97.27 (2)	95.2 (2)	69.6 (3)	72.2 (3)	R4=2.7

Figure 6 is observed closely, then it can be seen that the individual impact of a transition count-based feature and combination of topological feature with CG-based circle approaches is far less than the combination of these two. This indicates that the feature set produced by combining the transition count and CG-based circular approaches along with topological feature has the ability to recognize the online Bangla character sample with highest level of accuracy.

4.1 Statistical Significance Tests

Statistical significance tests have also been performed for validating the performance of the multiple classifiers using multiple datasets. Details of these tests can also be found in the previous work described in Reference [45]. In the present work, we have performed a non-parametric Friedman test along with a post-hoc test called Nemenyi test. The number of datasets (N) and the number of classifiers (k) are set as 10 and 4, respectively. Recognition accuracies attained by four different classifiers on 10 randomly selected datasets are shown in Table 5. On the basis of these results, the classifiers are ranked for each dataset, and the best-performing algorithm gets the rank 1, the second best classifier gets rank 2, and so on (see Table 5). In case of ties, average ranks are assigned. Let r_j^i be the rank of the j th classifier on i th dataset. Then, the mean of the ranks of the

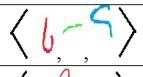
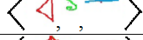
Character	Variation of writing style in terms of stroke number and shape	Number of strokes
		3
		2
		3
		2
		1

Fig. 7. Variations of different writing styles.

j th classifier over all the N datasets is computed as

$$R_j = \frac{1}{N} \sum_{i=1}^N r_j^i. \quad (1)$$

The null hypothesis states that all the classifiers are equivalent, and so their ranks R_j should be equal. The Friedman statistic [45] is computed as follows:

$$X_F^2 = \frac{12N}{k(k+1)} \left[\sum_j R_j^2 - \frac{k(k+1)^2}{4} \right]. \quad (2)$$

This statistic is distributed according to X_F^2 with $k-1(=3)$ degrees of freedom. Using Equation (2), the value of X_F^2 is calculated as 18.305. From the table of critical values, it can be observed that the value of X_F^2 with 3 degrees of freedom is 7.81 for $\alpha = 0.05$ (where α denotes the level of significance). It can be seen that the computed X_F^2 differs significantly from the standard X_F^2 . So the null hypothesis is rejected.

Since the null hypothesis is rejected, a post-hoc test known as the Nemenyi test is carried out for pairwise comparisons of the best- and worst-performing classifiers. Performances of two classifiers are significantly different if the corresponding average ranks differ by at least the critical difference (CD) which is expressed as

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6N}}. \quad (3)$$

For Nemenyi's test, the value of $q_{0.05}$ for the four classifiers is 2.569 (see Table 5(a) of Reference [12]). So the CD is calculated as $2.569 \sqrt{\frac{4.5}{6.10}}$, i.e., 1.483 using Equation (3). As the values of mean ranks between the best- (SVM) and worst- (BayesNet) performing classifiers and CD are significantly different, we can conclude that there is a substantial difference between the abilities of the SVM classifier with the others. Hence, after the application of statistical significance tests, it can be concluded that SVM classifier outperforms all other classifiers.

The experiment was performed on a 10,000-character database used by Sen et al. in previous experiments [37–40]. The SVM classifier accuracy is 98.7% when the value of *coef0* is 0 (default value). After tuning the value of *coef0*, the accuracy increases to 99.48% when *coef0* is set to 3, which is higher than the results from the past experiments done by Sen et al. Although the recognition accuracy falls slightly with an increase of the database size, this feature extraction strategy can be included as an effective technique towards recognition. The reduction in accuracy may be due to more writing variations (see Figure 7) that have been incorporated in the additional 5,000-character

Table 6. Some Misclassified Character Pairs (Experimented by SVM Classifier)

Mostly Misclassified character pairs	Misclassification (in %) out of 600 samples
(ঢ,ঢ়)	20 (3.3%)
(ম,স)	12 (2.0%)
(ত,ভ)	14 (2.3%)
(ই,ঈ)	12 (2.0%)
(ঋ,ঌ)	4 (0.7%)

Table 7. A Brief Summary of the Present Work along with Some Past Works

Reference	Used Features	Database Size	Accuracy (in %)
[9]	Directional Code Feature	7,043	83.61
[31]	Structural + Point Based Feature	12,500	91.13
[27]	Shape Based Feature (Shape and Size of stroke)	24,500	87.7
[30]	Structural + Point Based Feature	21,372	88.23
[19]	Structural and Directional Feature	4,900 (test samples)	87.48
[37]	CG Based Circle Approach (Global and Local Information)	10,000	98.26
[38]	Area Feature + Mass Distribution + Chord Length Feature	10,000	98.5
[39]	Distance Based Feature	10,000	98.2
[40]	Hausdorff Distance Based Feature	10,000	95.57
Present work	Combination of Transition Count + CG based Circular + Topological Feature	10,000	99.48
		15,000	95.49

dataset. Some misclassified character sample pairs are shown in Table 6. Observing the samples, it can be seen that the strong structural similarity and cursive nature of Bangla script are the main reasons for misclassification of those character samples.

4.2 Result Analysis

From Table 6, it can be seen that the most confusing pair of characters is characters ‘ঢ’ and ‘ঢ়’. Observing the characters carefully, we have noted that the characters bear similarity in every aspect barring the right most portion where the presence of a loop differentiates them. Character samples ‘ম’ and ‘স’ are similar to each other except that character ‘ম’ has a loop in the bottom left of the first stroke. Three more confusing character pairs are (‘ত’, ‘ভ’), (‘ই’, ‘ঈ’) and (‘ঋ’, ‘ঌ’) which are shown in the third, fourth, and fifth entries of the table, respectively.

Table 7 summarizes various information (volume of the database, feature set used, observed success rate) for the present work along with some other works available in the literature. Works described in References [37–40] have also been evaluated on the same 10,000 character database and it may be noted that the accuracy achieved in the present work is higher. Other entries of this table are given for quick review of some important researches on this topic. Some of the authors, as reported in Table 7, such as in References [19, 27, 30], have used stroke-based approaches while

Table 8. A Brief Comparison of the Present Work with Some Past Works over a 15,000-Character Database

Reference	Used Features	Database size	Computation Time(ms)	Accuracy (in %)
[31]	Structural + Point Based Feature	15,000	33.1	90.36
[9]	Directional Code Feature	15,000	45.47	90.39
[37]	CG Based Circle Approach (global and local information)	15,000	31.1	87.07
[38]	Area + Mass Distribution + Chord Length Feature	15,000	32.44	86.42
[39]	Distance Based	15,000	45.01	82.48
[40]	Hausdorff Distance Based Feature	15,000	32.73	79.15
Present work	Combination of Transition Count + CG based Circular + Topological Feature	15,000	32.2	95.49

others have considered the whole character image (holistic approach) for recognition. Hence, for comparison, we have only followed the approaches where attempts are made to recognize the characters holistically similar to our approach. We have evaluated the feature vectors on a 15,000-character database used in the present experiment and tested with the SVM classifier. The observed outcomes with feature computation time (in millisecond) for all the methods are shown in Table 8. The computational cost mentioned here indicates the observed average running time for feature extraction from a single character image. The experiments are done on a machine with an Intel processor with 2.16GHz processing speed and 4GB RAM. For the record, it can be said that the present technique outperforms all other previous techniques in terms of processing time as well as recognition accuracy.

5 CONCLUSIONS

In our work, we proposed a set of novel feature vectors designed for the development of a robust online handwritten Bangla character recognition system. The discriminatory nature of the features introduced here aided the classifiers in recognizing the large number of character classes. Hence, feature vectors produced satisfactory recognition accuracy on a reasonably large database of 15,000 handwritten online character samples of Bangla script. We tried to incorporate a wide variety of character samples during database preparation, as these variations can help assessing the robustness of any system. We observed some difficulties during recognition when different characters had almost identical writing pattern. Therefore, one of the possible future tasks would be to develop features to cope up with this misclassification. Additionally, we will look for some machine-learning approaches recently followed by the other researchers like deep learning, extreme learning, and so on.

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